

A7 8

11)
a)

$$\bar{X} = E(X) = \frac{a+b}{2} \rightarrow 2\bar{X} - a = b$$

$$a < b$$

$$\hookrightarrow a < 2\bar{X} - a$$

$$\hookrightarrow 2a < 2\bar{X}$$

$$\hookrightarrow a < \bar{X}$$

b) $a+b=0$

$$\bar{X} = E(X) = \frac{a+b}{2} \rightarrow \bar{X} = 0 \quad \text{Ne messen}$$

$$\overline{X^2} = E(X^2) = \int_a^b \frac{1}{b-a} X^2 \rightarrow \frac{X^3}{3} \cdot \frac{1}{(b-a)} \Big|_a^b$$

$$\frac{(b-a) \frac{b^3 - a^3}{3}}{3(b-a)}$$

$$= \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)}$$

$$= \frac{b^2 + ab + a^2}{3} = \frac{(b+a)^2 - ab}{3}$$

$$\boxed{\overline{X^2} = \frac{-ab}{3}} \rightarrow 3\overline{X^2} = -ab$$

$$\overline{X^3} = E(X^3) = \int_a^b \frac{1}{b-a} x^3 dx =$$

X^3 p.p.p.p.
no sign

$$\frac{x^4}{4(b-a)} \Big|_a^b = \frac{b^4 - a^4}{4(b-a)}$$

$$= \frac{(b^2 - a^2)(b^2 + a^2)}{4(b-a)}$$

$$\overline{X^4} = E(X^4) = \int_a^b \frac{1}{b-a} x^4 dx$$

$$= \frac{(b-a)(b+a)(b^2+a^2)}{4(b-a)} = 0$$

$$= \frac{b^5 - a^5}{5(b-a)}$$

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 1 & & 1 & & \\ & 1 & 2 & 1 & & & \\ 1 & 3 & 3 & 1 & & & \\ 1 & 4 & 6 & 4 & 1 & & \end{array}$$

$$= \frac{(b-a)(b^4 + a^3b + a^2b^2 + ab^3 + a^4)}{5(b-a)}$$

$$\overline{X^4} = \frac{(b+a)^4 - 3a^3b - 3a^2b^2 - 3ab^3}{5}$$

p2) $X_1, \dots, X_n \sim f(x|\sigma) \mid \begin{cases} \frac{2x}{\sigma^2} & 0 \leq x \leq \sigma \\ 0 & \text{e. o. c.} \end{cases}$

$$L(\sigma|X) = 2^{-n} \sigma^{-2n} \prod_{i=1}^n f(x_i) \quad \text{for } (0, \sigma)$$

$$\ln(L(\sigma|X)) = n \ln(2) - 2n \ln(\sigma) + \sum_{i=1}^n \ln(f(x_i)) \quad \left| \frac{d}{d\sigma} \right.$$

$$-\frac{2n}{\sigma} \text{ is decreasing.}$$

$\rightarrow \sigma$ deve ser o maior
pequeno possível

$$X_i \leq \sigma \rightarrow \boxed{\sigma_{\text{EMV}} = \max X_i} = Y$$

$$F_X(x) = \int_0^x \frac{2\bar{\sigma}}{\sigma^2} d\bar{\sigma} \rightarrow \left. \frac{\bar{\sigma}^2}{\sigma^2} \right|_0^x \rightarrow \begin{cases} 1 & x \geq \sigma \\ \frac{x^2}{\sigma^2} & 0 \leq x \leq \sigma \\ 0 & x < 0 \end{cases}$$

$$F_Y(\tau) = P(Y \leq \tau) = P(X_1 \leq \tau, \dots, X_n \leq \tau) = [F_X(\tau)]^n$$

$$= \left(\frac{\tau^2}{\sigma^2}\right)^n \quad \left| \frac{d}{d\tau} \right.$$

$$f_Y(\tau) = \frac{2n}{\sigma^{2n}} \tau^{2n-1} \quad 0 \leq \tau \leq \sigma$$

$$E(Y) = \int_0^\sigma \frac{2n}{\sigma^{2n}} \tau^{2n-1} \cdot \tau d\tau \rightarrow \frac{2n}{2n+1} \frac{1}{\sigma^{2n}} \cdot \sigma^{2n+1} \Rightarrow \frac{2n}{2n+1} \sigma$$

$$E\left(\frac{2n+1}{2n} Y\right) = \sigma$$

$$\hat{\sigma}_1 = \frac{2n+1}{2n} \max X_i$$

$$c) \text{Var}(\bar{O}^*) = \text{Var}\left(\frac{2m+1}{2m} Y\right) \\ = \left(\frac{2m+1}{2m}\right)^2 \text{Var}(Y)$$

$$\text{Var}(Y) = E(Y^2) - E(Y)^2$$

$$E(Y^2) = \int_0^{\sigma} \frac{2m}{\sigma^{2m}} \tau^{2m-1} \cdot \tau^2 d\tau = \frac{2m}{\sigma^{2m}} \frac{\sigma^{2m+2}}{2m+2} = \frac{m}{m+1} \sigma^2$$

$$\text{Var}(Y) = \frac{m}{m+1} \sigma^2 - \left(\frac{2m}{2m+1}\right)^2 \sigma^2 = \sigma^2 \left(\frac{m}{m+1} - \frac{4m^2}{(2m+1)^2} \right)$$

d) Case of interest

$$\begin{aligned} \text{ECM}(\bar{O}^*) &= \text{Var}(\bar{O}^*) - \text{Var}(\bar{O}^*) \\ &= \sigma^2 \left(\frac{m}{m+1} - \frac{4m^2}{(2m+1)^2} \right) \xrightarrow{m \rightarrow \infty} \\ &= \frac{\sigma^2 m}{(m+1)(2m+1)^2} \xrightarrow{\text{convert to order } O(n^{-2})} \frac{1}{1-1} \rightarrow 0 \quad \forall \sigma \end{aligned}$$

$$\frac{m}{m+1} - \frac{4m^2}{(2m+1)^2}$$

$$\frac{m}{m+1} - \frac{4m^2}{4m^2 + 4m + 1}$$

$$\frac{4m^3 + 4m^2 + m - 4m^3 - 4m^2}{(m+1)(2m+1)^2}$$

$$e) \bar{X} = E(X) = \int_0^{\sigma} \frac{2x \cdot x dx}{\sigma^2} = \frac{2}{3} \frac{\sigma^3}{\sigma^2} = \frac{2}{3} \sigma$$

$$\hat{\sigma}_{ML} = \frac{3}{2} \bar{X}$$

$$E(\hat{\sigma}_{ML}) = \frac{3}{2n} \sum E(X_i) = \frac{3}{2n} \cdot n \cdot \frac{2}{3} \sigma = \sigma$$

$$E(X^2) = \int_0^{\sigma} \frac{2x}{\sigma^2} x^2 dx$$

$$\frac{1}{2} \frac{\sigma^4}{\sigma^2} = \frac{\sigma^2}{2}$$

$$\text{Var}(\hat{\sigma}_{ML}) = \frac{9}{4n^2} \sum \text{Var}(X_i) = \frac{9}{4n} \text{Var}(X)$$

$$\text{Var}(X) = \frac{\sigma^2}{2} - \frac{4}{9} \sigma^2$$

$$= \frac{9}{4n} \frac{\sigma^2}{18} \rightarrow \frac{\sigma^2}{8n} \text{ Comp } \sim O(n^{-1})$$

more EMV

$$\sigma^2 \left(\frac{1}{18} \right)$$

p3) $X_1, \dots, X_n \sim P_{\text{Pois}}(\lambda)$

$$g(\lambda) = \lambda^2 \quad g'(\lambda) = 2\lambda$$

Sistema para $\bar{X}$ es $\overset{\text{SUF}}{\text{m.c.}} \text{ completo}$

$$E(\bar{X}) = \lambda$$

$$V_{\bar{X}}(\bar{X}) = \frac{\lambda}{n}$$

$$E(\bar{X}^2) = V_{\bar{X}}(\bar{X}) + E(\bar{X})^2$$

$$E(\bar{X}^2) = \frac{\lambda}{n} + \lambda^2$$

$$L(\lambda/x) = \frac{\lambda^{\sum x_i} e^{-\lambda n}}{n! \lambda_i}$$

$$\log(\lambda/x) = \sum x_i \log(\lambda) - \lambda n + c$$

$$= \sum x_i - n\lambda$$

$$= -\frac{\sum x_i}{\lambda^2}$$

$$E(\bar{X}^2 - \frac{\bar{X}^2}{n}) = \frac{\lambda}{n} + \lambda^2 - \frac{\lambda^2}{n} = \lambda^2$$

$$\bar{X}^2 - \frac{\bar{X}^2}{n} = T(x) \text{ estimador de } \lambda^2$$

Como es función de n estadística
Completa $\rightarrow$ EIVUM

$$I_X(\lambda) = -E\left(\frac{\partial}{\partial \lambda} \log(f(x/\lambda))\right)$$

$$= -E\left(-\frac{\sum x_i}{\lambda^2}\right) = \frac{1}{\lambda^2} \sum E(x_i) = \frac{n}{\lambda^2} \lambda = \frac{n}{\lambda}$$

Como con $\frac{n}{\lambda}$ se puede estimar λ

$$\frac{n}{\lambda^2} \rightarrow \sqrt{\frac{4\lambda^3}{n}}$$

$$V(\bar{X}^2 - \frac{\bar{X}}{n}) = V(\bar{X}(1 - \frac{1}{n}))$$

$$V(\bar{X}^2) + V(\frac{\bar{X}}{n}) - 2 \text{Cov}(\bar{X}^2, \frac{\bar{X}}{n})$$

$$E(\bar{X}^2 \cdot \frac{\bar{X}}{n}) - E(\bar{X}^2) E(\frac{\bar{X}}{n})$$

$$E(\frac{\bar{X}^3}{n}) -$$

Utilisez méthode de K

pg 14 $X_1 \dots X_{n+1} \sim \text{Bernoulli}$

$$h(p) = P(\sum_{i=1}^n X_i > X_{n+1} | p)$$

$$T(X_1 \dots X_{n+1}) = \begin{cases} 1 & \text{si } \sum_{i=1}^n X_i > X_{n+1} \\ 0 & \text{sinon} \end{cases}$$

$$h(p) = E(T) = 1 \cdot P(\sum_{i=1}^n X_i > X_{n+1}) + 0 \cdot P(\sum_{i=1}^n X_i \leq X_{n+1}) = h(p)$$

$$E(T | S) = \text{Res Blauw}$$

$$= P(\sum_{i=1}^n X_i > X_{n+1} | \sum_{i=1}^n X_i = c)$$

$$= \frac{P(\sum_{i=1}^n X_i > X_{n+1}, \sum_{i=1}^n X_i = c)}{P(\sum_{i=1}^n X_i = c)}$$

If $Y \geq 0$
 $Pr(A) = 0$
 $Y > 0$

$$\rightarrow P(\sum_{i=1}^n X_i > X_{n+1}, \sum_{i=1}^n X_i = c, X_{n+1} = 0) + P(\sum_{i=1}^n X_i > X_{n+1}, \sum_{i=1}^n X_i = c, X_{n+1} = 1)$$

$$Y=0 \quad P_r(\hat{\Sigma}^T X_i > 0, \hat{\Sigma}^T X_i = Y) = P_r(\Sigma^T X_i > 0) = \binom{n}{Y} p^Y (1-p)^{n-Y}$$

$$\text{if } Y=1 \text{ or } 2 \quad P_r(\hat{\Sigma}^T X_i > 1, \hat{\Sigma}^T X_i = Y-1) = 0$$

$$Y > 2 \quad P_r(\Sigma^T X_i > 1, \Sigma^T X_i = Y-1) = P_r(\hat{\Sigma}^T X_i = Y-1) = \binom{n}{Y-1} p^{Y-1} (1-p)^{n-Y+1}$$

jumlah di atas

$$E(T | \sum_{i=1}^n X_i = Y) = \begin{cases} C & Y=0 \\ \frac{\binom{n}{Y} p^Y (1-p)^{n-Y} (1-p)}{\binom{n+1}{Y} p^Y (1-p)^{n-Y+1}} & Y=1 \text{ or } Y \geq 2 \end{cases}$$

$$\cancel{\binom{n}{Y} p^Y (1-p)^{n-Y}}$$

$$\frac{\binom{n}{Y} + \binom{n}{Y-1} p^Y (1-p)^{n-Y+1}}{\cancel{\binom{n}{Y} p^Y (1-p)^{n-Y+1}}}$$

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